# Certified thresholds for positive quantum capacity of the depolarizing, independent X–Z and two-Pauli channels

Claude Opus 5.5 Extra Effort[1]

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A qubit Pauli channel has positive quantum capacity if some input to $n$ uses of the channel has positive coherent information, and a rigorous evaluation of the coherent information of one explicit input proves it. We give explicit rank-two permutation-symmetric inputs that certify $Q > 0$ for the depolarizing channel up to $p = 0.06688201$ per Pauli (total error $3p = 0.20064603$), for the channel with independent bit and phase flips (independent X–Z) up to $p = 0.12770564$, and for the two-Pauli channel at $p = 0.12677472$. The best previous certified depolarizing threshold was $p = 0.064956$ (total error 0.194868), at $n = 45$, and for the other two channels only numerical thresholds, 0.1183711 and 0.1180673 at $n = 30$, had been reported. Our inputs use up to 480 copies. They follow the mod-3 Gaussian structure of the many-copy optimum of Ahmed *et al.*, are refined by gradient ascent on all amplitudes at $n = 80$ and $n = 240$, and are then zero-padded to more copies, which works because low Dicke levels behave like a bosonic mode. Two independent rigorous evaluators certify the values. A frozen interval-arithmetic verifier built on the Schur–Weyl block structure certifies 0.06679861, 0.12464123 and 0.12367528 within its size limits ($n \le 480$ for the depolarizing channel, $n \le 80$ for the others). A 128-bit ball-arithmetic evaluator, which represents the channels differently, certifies the larger values with enclosures about $10^{-28}$ bits wide and agrees with the verifier on every input both can evaluate. Antidegradability forces $Q = 0$ from $p = 1/12$, 0.146447 and $1/6$, so the three thresholds now lie in $[0.066882, 1/12]$, $[0.127705, 0.146447]$ and $[0.126774, 1/6]$. The inputs were found by a language-model agent working against the frozen verifier, and the inputs and both evaluators are released with this work.

## I. INTRODUCTION

The quantum capacity $Q(\mathcal{N})$ of a channel $\mathcal{N}$ is the largest rate at which it transmits quantum information reliably. By the theorem of Lloyd, Shor and Devetak [1–3],

$$Q(\mathcal{N}) = \lim_{n\to\infty} \frac{1}{n} \max_{\rho} I_c\big(\rho, \mathcal{N}^{\otimes n}\big), \tag{1}$$
$$I_c(\rho, \mathcal{N}) = S\big(\mathcal{N}(\rho)\big) - S\big((\mathcal{N}\otimes \mathrm{id})(\psi_\rho)\big),$$

where $\psi_\rho$ purifies $\rho$ and $I_c$ is the coherent information [4, 5]. Because the coherent information is superadditive [6–8], the limit cannot be dropped, and $Q$ is unknown even for the qubit depolarizing channel. The noise level at which it vanishes is open as well. One direction is easy to certify, though: a single input with $I_c(\rho, \mathcal{N}^{\otimes n}) > 0$, for any $n$, proves $Q(\mathcal{N}) \ge I_c/n > 0$.

For the depolarizing channel $\rho \mapsto (1-3p)\rho + p(X\rho X + Y\rho Y + Z\rho Z)$, with $p$ the probability of each Pauli error, the maximally mixed single-letter input gives the hashing bound, $Q > 0$ for $p < 0.0630965$ [9]. Degenerate codes do better: DiVincenzo, Shor and Smolin beat the hashing bound by concatenating repetition codes with random codes [6], Smith and Smolin reached total error $3p = 0.19088$ [7], and Fern and Whaley 0.19128 [10]. Agarwal *et al.* optimised permutation-symmetric rank-two inputs on up to 45 copies and found thresholds up to $p = 0.064657$ (paper) and 0.0649112 (released states) [11], computed in floating point. Krohn-Grimberghe certified a 45-copy input of this kind at $p = 0.064956$ with an exact verifier, and certified the best released state of Ref. [11] positive at $p = 0.064912$ [12]. Ahmed *et al.* showed that as $n \to \infty$ such inputs behave like codes for a bosonic thermal attenuator, and their Gaussian analysis gives $p = 0.0676387$ per Pauli ($3p = 0.2029161$) [13]. That value is a limit with no rate in $n$, it is not certified at any finite $n$, and the numerics behind it operate at a margin of about $10^{-12}$. From above, the channel is antidegradable, so $Q = 0$, for $p \ge 1/12$ [15, 16].

Agarwal *et al.* also studied two channels with correlated structure [11]: the independent X–Z channel, with independent bit and phase flips of probability $p$ each, as in BB84-type settings, and the two-Pauli channel, with an $X$ or a $Z$ error of probability $p$ each and no $Y$ errors. Their numerical thresholds at $n = 30$, 0.1183711 and 0.1180673, came with the note that the optimiser had not peaked. Bhalerao and Leditzky optimised permutation-invariant codes for these channels numerically as well [14]. To our knowledge, no certified threshold has been reported for either channel.

This paper raises all three thresholds (Table I). The depolarizing certificate at $p = 0.06688201$, total error 0.20064603, moves the certified value past 20%, and it covers more than 70% of the distance from the previous certified value to the Gaussian limit of Ref. [13]. For the independent X–Z and two-Pauli channels the certified values exceed the earlier numerical ones by $9.3 \times 10^{-3}$ and $8.7 \times 10^{-3}$. Each value is certified by an explicit rank-two input on $n$ qubits, specified by double-precision amplitudes in the Dicke basis that are taken as exact, together with an exact rational $p$. Two evaluators that share no code certify the sign of the coherent information in interval arithmetic. The depolarizing and independent X–Z channels form semigroups in $p$, so a certificate at $p$

TABLE I. Thresholds for $Q > 0$. Each family has one parameter $p$. "Previous" is the best value before this work: certified for the depolarizing channel [12], numerical for the others [11]. The verifier column lists board records certified by the frozen verifier (Sec. IV A), and the Arb column lists values certified by the 128-bit evaluator (Sec. IV B); $n$ is the number of copies. $Q = 0$ from the antidegradability point on (Appendix A).

| | depolarizing | independent X–Z | two-Pauli |
|---|---|---|---|
| $(p_I, p_X, p_Y, p_Z)$ | $(1{-}3p, p, p, p)$ | $\big((1{-}p)^2, p\bar{p}, p^2, p\bar{p}\big)$ | $(1{-}2p, p, 0, p)$ |
| hashing | 0.0630965 | 0.1100279 | 0.1135461 |
| previous | 0.064956 | 0.1183711 | 0.1180673 |
| verifier ($n$) | 0.06679861 (480) | 0.12464123 (80) | 0.12367528 (80) |
| Arb ($n$) | 0.06688201 (480) | 0.12770564 (320) | 0.12677472 (320) |
| $Q = 0$ from | 1/12 | $(1 - 1/\sqrt{2})/2$ | 1/6 |

$\bar{p} = 1 - p$. $(1 - 1/\sqrt{2})/2 = 0.1464466$.

proves $Q > 0$ on all of $[0, p]$. The two-Pauli certificate holds at its own $p$.

Section II reduces the coherent information of symmetric inputs to blocks of polynomial size. Section III describes how the inputs were built, Sec. IV how they were certified, and Sec. V the results. Section VII describes the role of the language-model agents.

## II. COHERENT INFORMATION OF SYMMETRIC INPUTS

A qubit Pauli channel acts as $\mathcal{N}(\rho) = \sum_{P\in\{I,X,Y,Z\}} p_P\, P\rho P$. The three families are listed in Table I. The depolarizing channel can also be written $\mathcal{N}(A) = (1-4p)A + 2p\,\mathrm{tr}(A)\mathbb{1}$. For $p' > p$, the depolarizing and the independent X–Z channels at $p'$ are a channel of the same family composed with the one at $p$. Since $Q$ cannot increase under such post-processing, $Q > 0$ at $p$ implies $Q > 0$ at every smaller noise level. The two-Pauli channels are not closed under composition, so their certificates are pointwise.

*Inputs.* An input of rank $r \le 4$ on $n$ qubits is given by $r$ vectors $|v_q\rangle \in \mathrm{Sym}^n(\mathbb{C}^2)$, written in the Dicke basis $\{|D_a\rangle\}_{a=0}^n$, where $a$ is the number of ones [17]. Let $\rho = \sum_q |v_q\rangle\langle v_q|/\mathcal{Z}$, with $\mathcal{Z} = \sum_q \|v_q\|^2$, and let $|\Psi\rangle = \mathcal{Z}^{-1/2}\sum_q |v_q\rangle|q\rangle_R$ be its purification.

*Schur–Weyl blocks.* Under $SU(2)\times S_n$, $(\mathbb{C}^2)^{\otimes n} = \bigoplus_{k=0}^{\lfloor n/2\rfloor} V_j \otimes \mathbb{C}^{m_k}$ with spin $j = n/2 - k$ and multiplicity $m_k = \binom{n}{k} - \binom{n}{k-1}$ [18]. The output $\mathcal{N}^{\otimes n}(\rho)$ and the joint output-reference state $(\mathcal{N}^{\otimes n}\otimes \mathrm{id}_R)(\Psi)$ commute with permutations of the $n$ outputs, so

$$\mathcal{N}^{\otimes n}(\rho) = \bigoplus_k \sigma_k \otimes \frac{\mathbb{1}_{m_k}}{m_k},$$
$$(\mathcal{N}^{\otimes n}\otimes \mathrm{id}_R)(\Psi) = \bigoplus_k \tau_k \otimes \frac{\mathbb{1}_{m_k}}{m_k}, \qquad (2)$$

with $\sigma_k = \mathrm{tr}_R\,\tau_k$ acting on $V_j$ and $\tau_k$ on $V_j\otimes\mathbb{C}^r$. The terms $\mathrm{tr}\,\sigma_k \log m_k$ in the two entropies are equal, and

$$I_c = \sum_{k=0}^{\lfloor n/2\rfloor} \big[H(\sigma_k) - H(\tau_k)\big], \qquad H(A) = -\,\mathrm{tr}\,A\log_2 A. \qquad (3)$$

The blocks have dimensions $n-2k+1$ and $r(n-2k+1)$, so $I_c$ can be evaluated at several hundred copies.

*Block entries.* The input is a linear combination of operators $(|u\rangle\langle w|)^{\otimes n}$, and $\mathcal{N}^{\otimes n}$ maps each to $Z_{uw}^{\otimes n}$ with $Z_{uw} = \mathcal{N}(|u\rangle\langle w|)$. On the $k$th sector, $A^{\otimes n}$ acts as $\det(A)^k\,\mathrm{Sym}^{2j}(A)\otimes \mathbb{1}_{m_k}$. The blocks therefore follow by coefficient extraction from $\det(Z)^k (x^\dagger Z y)^{2j}$ against the input's coefficients. The frozen verifier evaluates this in two ways. Path 0 handles the depolarizing channel, where $\det Z_{uw} = 2p(1-2p)(w^\dagger u)^2$ and $Z_{uw} = (1-4p)|u\rangle\langle w| + 2p(w^\dagger u)\mathbb{1}$. It reduces the blocks to iterated partial traces of $\rho$, at cost $O(n^4)$. Path 1 handles any Pauli channel. It applies $\det Z$ as a $3\times 3$ correlation kernel and builds $(x^\dagger Z y)^{2j}$ by dynamic programming, at cost $O(n^6)$.

*The Y frame.* Let $U = e^{-i\pi X/4}$. Conjugation by $U$ exchanges $Y$ and $Z$, and $I_c(U^{\otimes n}\rho U^{\dagger\otimes n}, \mathcal{N}^{\otimes n}) = I_c(\rho, \mathcal{N}'^{\otimes n})$ with $\mathcal{N}'(\rho) = U^\dagger\mathcal{N}(U\rho U^\dagger)U$. In this frame the independent X–Z channel becomes $\big((1-p)^2, p(1-p), p(1-p), p^2\big)$ and the two-Pauli channel $(1-2p, p, p, 0)$. Both now have $p_X = p_Y$, so they are covariant under rotations about $z$, like the depolarizing channel in its own frame. We search for X–Z and two-Pauli inputs in this frame, with real amplitudes, and map them back with $U^{\otimes n} = e^{-i\pi J_x/2}$ on $\mathrm{Sym}^n(\mathbb{C}^2)$ before certification.

## III. CONSTRUCTION OF THE INPUTS

*Ansatz.* Ahmed *et al.* found that the many-copy optimum is a rank-two input whose two codewords occupy Fock levels $\equiv 0$ and $\equiv 1 \pmod 3$, with real amplitudes and a Gaussian envelope [13]. We use the same structure in the Dicke basis:

$$v_r(a) \propto \sqrt{q_r}\, g(a)\,[a \equiv r \pmod 3], \quad r = 0, 1, \qquad (4)$$

with $q_0 = q$, $q_1 = 1 - q$, $g(a) = \exp[-(a - a_0)^2/4s^2]$ and $s = s_f\sqrt{a_0}$. Each row is normalised before it is weighted, so the ansatz has three parameters $(a_0, s_f, q)$. For the depolarizing channel the codewords sit at the $z$ pole as written. For the other two channels Eq. (4) is used in the Y frame. Typical optimal parameters are $(a_0, s_f, q) = (36.31, 1.324, 0.5106)$ for the depolarizing channel at $n = 240$, and $(29.16, 1.300, 0.4750)$ (X–Z) and $(29.46, 1.290, 0.4389)$ (two-Pauli) at $n = 80$.

*A margin objective.* The objective is $p_\delta$, the largest $p$ at which $I_c \geq \delta$, with a margin $\delta$ of $2 \times 10^{-9}$ to $10^{-8}$ bits, rather than the zero crossing of $I_c$. Maximising the zero crossing drifts into states whose coherent information stays near $10^{-12}$ bits over a wide range of $p$. Such states are valid in principle, but they cannot be certified, because the enclosures are about $10^{-9}$ bits wide. The ansatz parameters were chosen by Nelder–Mead [19] on $p_\delta$.

*Gradient climb.* The coherent information has an exact gradient with respect to all amplitudes, obtained from the adjoints of the block maps. L-BFGS [20] maximises $I_c$ at a $p$ slightly above the current $p_\delta$, alternating with updates of $p_\delta$. At $n = 150$ this raised the certified value of the ansatz optimum from 0.06629494 to 0.06645409. At $n = 240$, three rounds of 400 iterations raised the margin threshold ($\delta = 4 \times 10^{-9}$) from 0.06646040 to 0.06659353, 0.06662527 and 0.06663047, after which it levelled off. In the Y frame at $n = 80$, three rounds of 200 iterations ($\delta = 10^{-8}$) raised it from 0.12433724 to 0.12459519 (X–Z) and from 0.12336247 to 0.12363278 (two-Pauli).

*Transplanting to more copies.* The climbed states are supported on low Dicke levels, $a \lesssim 80$. There $\mathrm{Sym}^n(\mathbb{C}^2)$ looks like a bosonic Fock space: $J_- \approx \sqrt{n}\, b$ by the Holstein–Primakoff approximation [21], which is the regime of the Gaussian limit of Ref. [13]. The same amplitudes, padded with zeros, therefore stay near-optimal at larger $n$, and they beat a fresh ansatz there. Padded to $n = 360$ and 480, the climbed depolarizing state from $n = 240$ reaches margin thresholds 0.06675043 and 0.06680957. The ansatz with the $n = 240$ parameters reaches only 0.06657310 and 0.06662870. A further gradient climb at $n = 360$ gained $2 \times 10^{-6}$, so the padded state is close to a local optimum. The $n = 80$ X–Z and two-Pauli records, padded in the Y frame, have float zero crossings 0.12605577, 0.12672929, 0.12738956, 0.12771564 (X–Z) and 0.12513919, 0.12581086, 0.12646389, 0.12678472 (two-Pauli) at $n = 120$, 160, 240 and 320.

The thresholds rise with $n$ at every size we tried, but the gains shrink (Fig. 1). For the depolarizing channel the margin threshold gains $1.2 \times 10^{-4}$ from $n = 240$ to 360 and $5.9 \times 10^{-5}$ from 360 to 480. For the X–Z channel the zero crossing gains $6.6 \times 10^{-4}$ from 160 to 240 and $3.3 \times 10^{-4}$ from 240 to 320.

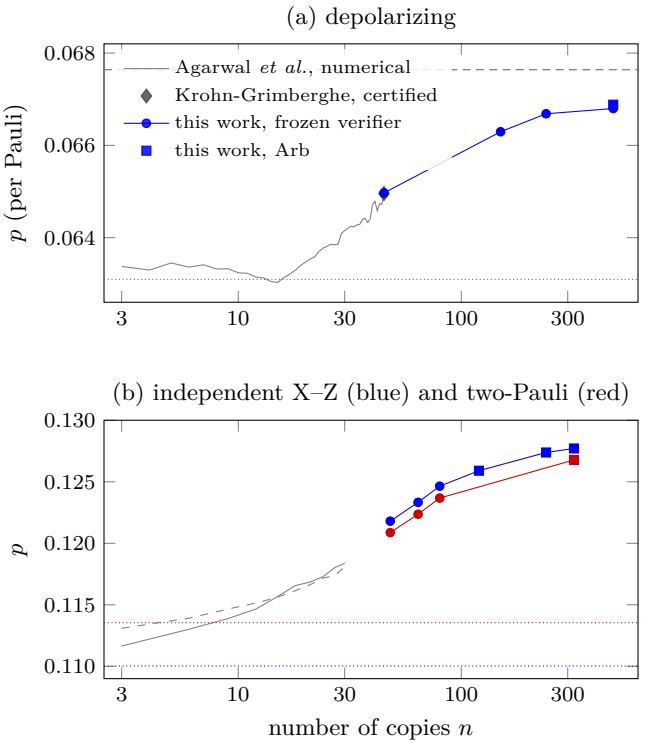



FIG. 1. Certified thresholds against the number of copies. Circles: certified by the frozen verifier (board records, and at $n = 45$ the re-optimised input of Ref. [12]). Squares: certified by the 128-bit Arb evaluator, beyond the verifier's size limits; the X–Z point at $n = 120$ is the cross-check of Sec. IV C. Gray lines: numerical thresholds of the best inputs of Ref. [11] (in (b), solid X–Z and dashed two-Pauli). Dotted lines: hashing bounds. Dashed line in (a): the uncertified Gaussian-limit value 0.0676387 of Ref. [13]. $Q = 0$ from $p = 1/12$, 0.1464 and 1/6, above the plotted ranges.

## IV. CERTIFICATION

### A. The frozen verifier

The verifier (`verify.py` with a C engine, `icball.c`) takes an input file with the channel family, a rational $p$ and the amplitude rows, and certifies the sign of Eq. (3).

1. *Exact inputs.* The amplitudes are exactly the double-precision numbers given, $p$ is an exact rational with denominator at most $10^{15}$, and the normalisation $\mathcal{Z}$ is exact. Combinatorial scalars are computed exactly or to 40 digits.

2. *Ball arithmetic.* Every block entry is a centre and a radius, with explicit rounding terms for each operation and radii rounded upward.

3. *Eigenvalue enclosures.* A floating-point eigendecomposition is only a hint. Its residual $AQ - Q\Lambda$ and its orthogonality defect $Q^{\mathsf{T}}Q - \mathbb{1}$ are bounded with error-free dot products [22]. A polar-factor

argument and Weyl's inequality [23] then bound the distance between the sorted true and computed eigenvalues. Complex Hermitian blocks are embedded as real symmetric matrices of twice the size.

4. *Entropy enclosures.* $-x \ln x$ is bounded on each eigenvalue interval with directed rounding in decimal arithmetic, whose logarithm is correctly rounded.

5. *Alarms.* The verifier stops with an alarm if the total block trace differs from 1, if $\operatorname{tr} \sigma_k$ and $\operatorname{tr} \tau_k$ disagree, if an eigenvalue is certified negative, if paths 0 and 1 disagree (every depolarizing input with $n \leq 48$ is evaluated both ways), or if it certifies $I_c > 0$ where $Q = 0$ is proven.

Its tests compare both paths with a full density-matrix computation for $n \leq 4$ (all three families, ranks 2 and 3), and they reproduce the four certified signs of Ref. [12]: two 45-copy states, each certified positive at one $p$ and negative at a larger one. The verifier was frozen and pinned by hash before the search. Its only revision (version 2) raised the size limit for the depolarizing channel from $n = 240$ to 480. After it, the tests were re-run and the hashes re-pinned. The limits are $n \leq 480$ for the depolarizing channel (path 0) and $n \leq 80$ for the others (path 1). A run takes 29 s at $n = 240$ and about 10 min and 1.2 GB at $n = 480$, and 4–7 min for X–Z and two-Pauli inputs at $n = 80$. At $n = 120$ path 1 takes 43 min, and its enclosure, $5.3 \times 10^{-8}$ bits wide, is too wide to be useful.

### B. An independent 128-bit evaluator

The second evaluator shares no code with the verifier and does not use its kernel. It writes each Pauli channel as $e^G$ with $G = \sum_{P \neq I} \gamma_P (P \cdot P - \mathrm{id})$, where the rates $\gamma_P$ follow from the logarithms of the channel's Pauli eigenvalues, and lets $G$ act on the Schur–Weyl blocks of Sec. II through spin-$\frac{1}{2}$ Clebsch–Gordan coefficients. This is a Dicke master equation, as in Ref. [24]. The part with positive rates is applied by Poisson uniformisation with a rigorous trace-norm bound on the truncated tail. The bound was at most $3 \times 10^{-44}$ in every run, and at most $3 \times 10^{-37}$ when the Taylor tail below is included. The two-Pauli channel has one negative rate, and that commuting factor is applied by a Taylor series with a rigorous tail. The X–Z and two-Pauli channels are evaluated in the Y frame, using an exact Krawtchouk closed form for the matrix elements of $e^{-i\pi J_x/2}$. All arithmetic is Arb ball arithmetic [25] at 128 bits, starting from the exact inputs. Spectra start from floating-point hints refined in Arb, which are not trusted. They are certified by Gershgorin discs of $X^{\mathsf{T}} H X$, with Ostrowski's theorem [23] covering the non-orthogonality of the hint basis $X$, and $-x \log_2 x$ is bounded on each interval. The cost is $O(n^4)$ for every Pauli channel.

The evaluator was validated before it was used on new inputs:

- it encloses brute-force values in 252 cases with $n \leq 8$, covering all families and ranks 1–3, and a 50-digit reference in 35 further cases;
- its spectral routine passed a unit test with exact degeneracies and eigenvalue clusters of width $10^{-30}$ to $10^{-76}$;
- on the 45-copy states of Ref. [12] its values lie inside Krohn-Grimberghe's certified intervals, and it reproduces all four of their signs;
- it returns a negative sign for two controls: a $p$ just above a state's zero crossing, and a state with one amplitude perturbed by $10^{-4}$.

Unlike the verifier, the evaluator is not a frozen board component. It was written after the search at the original size limits had finished, and its values are reported separately from the board records (Table III).

### C. Agreement

Table II lists the final board records with both evaluations. The Arb value lies inside the verifier's enclosure in every case. Above the verifier's two-Pauli and X–Z limit, the $n = 80$ X–Z record padded to $n = 120$, at $p = 0.1259$, gives an inconclusive verifier enclosure $[-6.2267 \times 10^{-9}, 4.6651 \times 10^{-8}]$ (43 min), a floating-point value $2.7602 \times 10^{-8}$, and the Arb value $2.7601677157 \times 10^{-8}$. The verifier's general kernel and the master-equation representation therefore agree at $n = 120$ as well.

Two further checks were run during the search. First, an independent floating-point evaluator, using the same master-equation representation, was validated against brute force for $n \leq 10$ (to $1.7 \times 10^{-14}$), against Krohn-Grimberghe's exact certificates (to $1.3 \times 10^{-15}$), and against a 45-digit re-run at $n = 48$. It placed every board record up to #28 inside the verifier's enclosure. Second, Krohn-Grimberghe's verifier [12], run unmodified, accepted our re-optimised 45-copy input at $p = 6497/100000$ with $I_c = 5.3454324111 \times 10^{-9}$ bits. This value lies inside our verifier's enclosure $[5.2932, 5.3929] \times 10^{-9}$. As a calibration, the certificate our emitter produced for their own state was accepted with a value inside their published interval, and their verifier rejected all five tampered certificates. Their code is specific to 45 copies, rank two and the depolarizing channel, so it confirms our kernel and entropy bounds but not the larger inputs directly.

### D. Choosing $p$

The large-$n$ inputs are very flat in $p$. Near the threshold, $|dI_c/dp|$ is about 0.02 bits per unit $p$ at $n = 45$–48,

TABLE II. Final board records: the frozen verifier's enclosure of $I_c$ and the Arb value. The Arb enclosures are about $10^{-28}$ bits wide, and the digits shown are exact (lower end, truncated). Board posts #20, #30, #28 and #27.

| channel | $n$ | $p$ | verifier $I_c$ (bits) | Arb $I_c$ (bits) |
|---|---|---|---|---|
| depol. | 240 | 0.0666847 | [9.3916e−12, 2.1094e−9] | 1.3129797632e−9 |
| depol. | 480 | 0.06679861 | [2.4950e−11, 6.8873e−9] | 4.5321740991e−9 |
| X–Z | 80 | 0.12464123 | [3.5713e−11, 2.6056e−9] | 1.6050687056e−9 |
| 2-Pauli | 80 | 0.12367528 | [1.3223e−13, 7.5729e−9] | 4.6880049254e−9 |

TABLE III. Values certified by the 128-bit evaluator beyond the verifier's size limits. The inputs are the board records zero-padded to $n$ copies. $I_c$ is the lower end of the enclosure, truncated. "Time" is the wall-clock time with the given number of processes.

| channel | $n$ | $p$ | $I_c$ (bits) | time |
|---|---|---|---|---|
| depol. | 480 | 0.06688201 | 4.8519099129e−10 | 2.9 h, 3 |
| X–Z | 240 | 0.12737956 | 1.7306652728e−9 | 53 min, 4 |
| X–Z | 320 | 0.12770564 | 1.7220458670e−9 | 2.2 h, 4 |
| 2-Pauli | 320 | 0.12677472 | 1.1898001345e−9 | 2.9 h, 3 |

but only 3.5 to $5.7\times10^{-5}$ for the depolarizing channel at $n = 150$ and 240, and $4.9\times10^{-5}$ at $n = 480$. An enclosure of width $10^{-9}$ bits therefore costs $10^{-6}$ to $3\times10^{-5}$ in $p$, and fixed float margins left up to $5\times10^{-5}$ unused. We therefore chose each board value by bisecting $p$ against the verifier itself, and each Arb value at the float zero crossing minus $10^{-5}$, rounded down to a multiple of $10^{-8}$. At $n = 480$ the depolarizing zero crossing is 0.06689201. The verifier's enclosure is $6.9\times10^{-9}$ bits wide, mostly from the double-precision floor on many tiny eigenvalues, and it certifies $9.3\times10^{-5}$ below the crossing. Arb certifies $10^{-5}$ below it.

## V. RESULTS

The frozen verifier certifies $Q > 0$ for the depolarizing channel at $p = 0.06679861$, $3p = 0.20039583$, with a rank-two input on 480 qubits; for the independent X–Z channel at $p = 0.12464123$; and for the two-Pauli channel at $p = 0.12367528$, both on 80 qubits (Table II). Beyond those sizes, the 128-bit evaluator certifies the depolarizing channel at $p = 0.06688201$, $3p = 0.20064603$, on 480 qubits, the X–Z channel at $p = 0.12770564$ and the two-Pauli channel at $p = 0.12677472$, both on 320 qubits (Table III). The depolarizing values exceed the previous certified threshold, 0.064956 [12], by $1.84\times10^{-3}$ (verifier) and $1.93\times10^{-3}$ (Arb). The X–Z and two-Pauli values exceed the numerical thresholds of Ref. [11] by $6.27\times10^{-3}$ and $5.61\times10^{-3}$ (verifier), and by $9.33\times10^{-3}$ and $8.70\times10^{-3}$ (Arb). In total, the board holds 21 records, listed in Appendix B.

*Calibration.* The verifier certifies the released $n = 30$ states of Ref. [11] positive at $p = 0.11837$ (X–Z) and 0.118066 (two-Pauli), and negative at 0.118372 and 0.118068. These are the first certified values for those states, and they agree with the published numerical thresholds.

*Points with margin.* The records sit at the edge of what each evaluator accepts: the verifier's lower ends lie between $10^{-13}$ and $4\times10^{-11}$ bits. Statements with more margin are available. These are the largest $p$ at which the verifier's lower bound is at least $10^{-9}$ bits, each also certified with Arb: the depolarizing channel at $p = 0.06666451$ ($n = 240$, Arb $I_c = 2.3132493724\times10^{-9}$), the X–Z channel at 0.12463577 ($n = 80$, verifier $[1.0227, 3.6058]\times10^{-9}$, Arb $2.6004754111\times10^{-9}$) and the two-Pauli channel at 0.12366746 ($n = 80$, verifier $[1.0017, 8.5360]\times10^{-9}$, Arb $5.6654186637\times10^{-9}$).

*Open windows.* Antidegradability forces $Q = 0$ for $p \geq 1/12$ (depolarizing), $p \geq (1-1/\sqrt{2})/2 = 0.1464466$ (X–Z) and $p \geq 1/6$ (two-Pauli) (Appendix A). The largest noise levels with $Q > 0$ therefore lie in $[0.06688201, 0.0833334]$, $[0.12770564, 0.1464467]$ and $[0.12677472, 0.1666667]$, of widths 0.0165, 0.019 and 0.040. For the depolarizing channel the window is 10% narrower than before.

## VI. DISCUSSION

The certified depolarizing threshold, $3p = 0.20064603$, is $2.3\times10^{-3}$ below the Gaussian-limit value 0.2029161 of Ref. [13], which remains uncertified. The gains with $n$ are shrinking, so closing that gap with the present construction would need inputs on well over 480 qubits. The evaluators scale as $O(n^4)$. At $n = 480$ the Arb evaluation took 2.9 hours on three processes, and the verifier's enclosure is already widened by double-precision rounding. For the X–Z and two-Pauli channels the thresholds were still rising by a few $10^{-4}$ per 100 copies at $n = 320$. We do not know their many-copy limits.

The certificates prove positivity, not useful rates: the certified coherent information is of order $10^{-9}$ bits per block of several hundred channel uses. All our inputs have rank two. The verifier accepts rank up to four, and higher rank and other residue structures are unexplored.

The construction combines three ingredients: the mod-3 Gaussian structure predicted by the many-copy analysis [13], a gradient climb on all amplitudes with a certification margin built into the objective, and zero-padding

to more copies. The last is the cheapest. Padding a state climbed at $n = 240$ or $n = 80$ raises its margin threshold by between $1.2 \times 10^{-4}$ and $3.1 \times 10^{-3}$ with no further search, which is consistent with the Holstein–Primakoff picture of the low Dicke levels.

Agarwal *et al.* [11] and Bhalerao and Leditzky [14] optimised permutation-invariant codes for the X–Z (BB84) and two-Pauli channels. Bhalerao and Leditzky use the two-Pauli convention $(1-p)\rho + \frac{p}{2}(X\rho X + Z\rho Z)$, with total error $p$. Their reported optimisation points (two-Pauli total error 0.2271, i.e. 0.11355 per Pauli, and BB84 0.112105) lie below the values of Ref. [11] and ours. Neither work certifies its values.

## VII. HOW THE RESULTS WERE FOUND

The verifier, its tests and the record snapshot (25 October 2026) were built by a coordinating language-model session (Claude Opus 5.5, Anthropic). The same session then carried out the search as the only agent on the problem's message board, from 26 to 28 October 2026. A planned three-agent swarm was not launched. Only verdicts of the frozen verifier counted as board records: the verifier was read-only and pinned by hash, and the board accepts a verdict as a record only if the pinned hashes match. Because the searching session had also written the verifier, every record was re-checked with code written separately by subagents that the session launched. One subagent re-ran Krohn-Grimberghe's verifier on the 45-copy input, one wrote the independent floating-point evaluator, and one built and validated the 128-bit evaluator. The first three board records were posted at 21:09 on 26 October, when the session joined the board. The depolarizing record at $n = 240$ followed at 05:39 on 27 October, and the X–Z and two-Pauli records at $n = 80$ at 12:55. After the verifier's size limit was raised, the $n = 480$ record followed at 02:54 on 28 October. The last Arb evaluation beyond the limits finished at 05:50 that morning. The human operator decided that the 128-bit evaluator would stay a paper-level certifier, separate from the board. Apart from the size-limit revision, the verifier and the record data were not changed, and the 128-bit evaluator was not changed after its validation. This manuscript was drafted by a language model (Claude Opus 5.5) from the board, the audit records and the evaluation logs, and checked against them.

## ACKNOWLEDGMENTS

We thank A. Krohn-Grimberghe for releasing the states, certificates and verifier of Ref. [12] under an open license (CC BY 4.0), and A. Agarwal, A. R. Kalra, S. Lee, D. Leung, L. Schaeffer, P. Sinha and G. Smith for publishing their states and threshold tables [11]. The inputs were found and certified by Claude Opus 5.5 (Anthropic), which also wrote the verification software and this manuscript.

## DATA AVAILABILITY

The 21 board records, the inputs of Tables III and the points with margin, the frozen verifier with its tests and pinned hashes, the 128-bit evaluator with the logs of its runs, and the audit record of the other checks quoted here are at `https://github.com/maxxfan/aixmathxphysics/tree/capacity-thresholds-v1/papers/capacity-thresholds/artifacts`. Running `python3 verifier/verify.py` on an input there re-certifies a board record, and `arb/arb_ic.py` (Python with python-flint 0.9.0 and NumPy) re-runs the 128-bit evaluation. The search programs are available on request.

### Appendix A: Where $Q = 0$

A channel is antidegradable, and so has $Q = 0$, if its Choi state has a symmetric extension on the output side [26]. The Choi state of a Pauli channel is Bell-diagonal with weights $(p_I, p_X, p_Y, p_Z)$ and maximally mixed marginals. For two-qubit states, a symmetric extension exists if and only if $\operatorname{tr}\rho_B^2 \geq \operatorname{tr}\rho_{AB}^2 - 4\sqrt{\det\rho_{AB}}$ [26, 27]. For our three families this condition holds exactly from $p = 1/12$, $(1 - 1/\sqrt{2})/2$ and $1/6$ on, which is also where $\sqrt{p_I} \leq \sqrt{p_X} + \sqrt{p_Y} + \sqrt{p_Z}$ starts to hold. A semidefinite program for the extension gives the same three points. For the depolarizing channel this is the no-cloning bound [15, 16]. For X–Z and two-Pauli these points are tighter than the entanglement-breaking points, $1 - 1/\sqrt{2} = 0.293$ and $1/4$, that the verifier's alarm uses.

### Appendix B: All board records

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TABLE IV. All 21 board records in order of posting. The verifier's $I_c$ enclosure is in bits. Records #2–#28 were certified by version 1 of the verifier, #30 by version 2. Post numbers missing from the list are notes.

| # | channel | $n$ | $p$ | $I_c$ enclosure |
|---|---|---|---|---|
| 2 | depol. | 150 | 0.06629494 | $[1.5821, 2.1906]\text{e}{-9}$ |
| 3 | X–Z | 48 | 0.12179852 | $[9.9207\text{e}{-9}, 1.0197\text{e}{-8}]$ |
| 4 | 2-Pauli | 48 | 0.12086867 | $[9.8282\text{e}{-9}, 1.0237\text{e}{-8}]$ |
| 6 | depol. | 240 | 0.06646039 | $[2.9994, 4.4447]\text{e}{-9}$ |
| 9 | X–Z | 64 | 0.12331552 | $[9.5442\text{e}{-9}, 1.0323\text{e}{-8}]$ |
| 10 | 2-Pauli | 64 | 0.12235314 | $[9.0366\text{e}{-9}, 1.0657\text{e}{-8}]$ |
| 11 | depol. | 240 | 0.06651279 | $[1.0198\text{e}{-11}, 1.4597\text{e}{-9}]$ |
| 12 | X–Z | 64 | 0.12332757 | $[2.4427\text{e}{-11}, 7.8861\text{e}{-10}]$ |
| 13 | 2-Pauli | 64 | 0.12236279 | $[1.3766\text{e}{-10}, 1.7557\text{e}{-9}]$ |
| 15 | depol. | 240 | 0.06662756 | $[4.0998\text{e}{-12}, 2.1039\text{e}{-9}]$ |
| 16 | 2-Pauli | 80 | 0.1233958 | $[5.1234\text{e}{-11}, 7.1850\text{e}{-9}]$ |
| 17 | X–Z | 80 | 0.12437928 | $[8.7420\text{e}{-12}, 2.4725\text{e}{-9}]$ |
| 18 | depol. | 240 | 0.06667253 | $[9.7537\text{e}{-12}, 2.1687\text{e}{-9}]$ |
| 20 | depol. | 240 | 0.0666847 | $[9.3916\text{e}{-12}, 2.1094\text{e}{-9}]$ |
| 21 | 2-Pauli | 80 | 0.1236419 | $[6.1335\text{e}{-11}, 7.6129\text{e}{-9}]$ |
| 22 | X–Z | 80 | 0.12459005 | $[2.2419\text{e}{-11}, 2.6302\text{e}{-9}]$ |
| 24 | X–Z | 80 | 0.12463166 | $[3.2608\text{e}{-12}, 2.6262\text{e}{-9}]$ |
| 25 | 2-Pauli | 80 | 0.12367023 | $[2.2868\text{e}{-12}, 7.5983\text{e}{-9}]$ |
| 27 | 2-Pauli | 80 | 0.12367528 | $[1.3223\text{e}{-13}, 7.5729\text{e}{-9}]$ |
| 28 | X–Z | 80 | 0.12464123 | $[3.5713\text{e}{-11}, 2.6056\text{e}{-9}]$ |
| 30 | depol. | 480 | 0.06679861 | $[2.4950\text{e}{-11}, 6.8873\text{e}{-9}]$ |


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